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PYTHON

Interview Questions



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Topic 1

Basic data types in Python:

Q.1 What are the basic data types in Python?

Ans: The basic data types in Python are:

- **Integer:** represents whole numbers.
- **Float:** represents decimal numbers.
- **String:** represents a sequence of characters.
- **Boolean:** represents either True or False.
- **List:** represents an ordered collection of elements.

Q.2 How do you convert a string to an integer in Python?

Ans: You can use the int() function to convert a string to an integer. For example:

```
num_str = "10"
num_int = int(num_str)
```

Q.3 How do you check the data type of a variable in Python?

Ans: You can use the **‘type()’** function to check the data type of a variable. For example:

```
num = 10
print(type(num))  # Output: <class 'int'>
```

Q.4 What is the difference between a list and a tuple in Python?

Ans: A list is mutable, which means you can modify its elements, while a tuple is immutable, meaning its elements cannot be changed after creation.

Q.5 How do you create an empty dictionary in Python?

Ans: You can create an empty dictionary using either the curly **braces {}** or the **dict()** function. For example:

```
empty_dict = {}
empty_dict = dict()
```

Topic 2

OOPS concept in Python:

Q.1 What is OOPS and how is it implemented in Python?

Ans: Object-Oriented Programming (OOPS) is a programming paradigm that uses objects to represent real-world entities. In Python, OOPS is implemented through classes and objects. Classes are blueprints for creating objects, and objects are instances of a class.

Q.2 What are the four principles of OOPS?

Ans: The four principles of OOPS are:

- **Encapsulation:** bundling of data and methods that operate on that data within a single unit (class).
- **Inheritance:** ability of a class to inherit properties and methods from its parent class.
- **Polymorphism:** ability of an object to take on different forms or behaviors based on the context.
- **Abstraction:** representing essential features and hiding unnecessary details to simplify the complexity.

Q.3 What is method overloading in Python?

Ans: Method overloading in Python refers to defining multiple methods with the same name but different parameters within a class. However, Python does not support method overloading by default as it does in languages like Java. In Python, you can achieve a similar effect by using default arguments or using variable-length arguments.

Q.4 What is method overriding in Python?

Ans: Method overriding in Python refers to defining a method in a child class that already exists in its parent class with the same name and signature. The method in the child class overrides the method in the parent class, providing a different implementation.

Q.5 What is the difference between a class method and an instance method in Python?

Ans: A class method is a method bound to the class and not the instance of the class. It is defined using the **`@classmethod`** decorator and can access only class-level variables. On the other hand, an instance method is bound to the instance of the class and can access both instance and class-level variables.

Topic 3

String handling functions:

Q.1 How do you concatenate two strings in Python?

Ans: You can concatenate two strings using the **+** operator. For example:

```
str1 = "Hello"
str2 = "World"
result = str1 + str2  # Output: "HelloWorld"
```

Q.2 How do you find the length of a string in Python?

Ans: You can use the **`len()`** function to find the length of a string. For example:

```
str1 = "Hello World"
length = len(str1)  # Output: 11
```

Q.3 How do you convert a string to uppercase in Python?

Ans: You can use the **`upper()`** method to convert a string to uppercase. For example:

```
str1 = "hello"
uppercase_str = str1.upper()  # Output: "HELLO"
```

Q.4 How do you split a string into a list of substrings in Python?

Ans: You can use the split() method to split a string into a list of substrings based on a delimiter. For example:

```
str1 = "Hello,World"
substrings = str1.split(",")
# Output: ["Hello", "World"]
```

Q.5 How do you check if a string contains a specific substring in Python?

Ans: You can use the **`in`** keyword to check if a substring is present in a string. For example:

```
str1 = "Hello World"
is_present = "World" in str1  # Output: True
```

Topic 4

Control statements, functions in Python:

Q.1 What are control statements in Python?

Ans: Control statements are used to control the flow of execution in a program. Common control statements in Python include if-else, for loops, while loops, and break/continue statements.

Q.2 How do you write an if-else statement in Python?

Ans: An if-else statement in Python is written using the following syntax:

```
if condition:
    # Code block executed if the condition is
True
else:
    # Code block executed if the condition is
False
```

Q.3 How do you define a function in Python?

Ans: A function in Python is defined using the **`def`** keyword. For example:

```
def greet():
    print("Hello, world!")
```

Q.4 How do you pass arguments to a function in Python?

Ans: You can pass arguments to a function by including them inside the parentheses when defining the function. For example:

```
def greet(name):
    print("Hello, " + name + "!")
```

Q.5 How do you return a value from a function in Python?

Ans: You can use the return keyword to return a value from a function. For example:

```
def add(a, b):
    return a + b
```

Topic 5

Special data types in Python:

Q.1 What is a set in Python?

Ans: A set in Python is an unordered collection of unique elements. It is defined using curly braces **{}** or the **set()** constructor. For example:

```
my_set = {1, 2, 3}  # Output: {1, 2, 3}
```

Q.2 What is a dictionary in Python?

Ans: A dictionary in Python is an unordered collection of key-value pairs. It is defined using curly braces **{}** or the **dict()** constructor. For example:

```
my_dict = {"name": "John", "age": 25}
# Output: {"name": "John", "age": 25}
```

Q.3 How do you access values in a dictionary in Python?

Ans: You can access values in a dictionary by using the corresponding key. For example:

```python
my_dict = {"name": "John", "age": 25}
print(my_dict["name"])  # Output: "John"
```

Q.4 What is a tuple in Python?

Ans: A tuple in Python is an ordered and immutable collection of elements. It is defined using parentheses `()` or the **`tuple()`** constructor. For example:

```python
my_tuple = (1, 2, 3)  # Output: (1, 2, 3)
```

Q.5 How do you swap the values of two variables in Python?

Ans: You can swap the values of two variables using a temporary variable or simultaneous assignment. For example:

```python
a = 5
b = 10
a, b = b, a
print(a, b)  # Output: 10, 5
```

Topic 6

Lambda functions, list comprehension:

Q.1 What is a lambda function in Python?

Ans: A lambda function is an anonymous function defined using the **`lambda`** keyword. It is typically used for short, one-line functions. For example:

```
square = lambda x: x**2
print(square(3))  # Output: 9
```

Q.2 What is list comprehension in Python?

Ans: List comprehension is a concise way to create lists in Python based on existing lists or other iterables. It combines the creation of a new list with a loop and optional conditional statements. For example:

```
numbers = [1, 2, 3, 4, 5]
squared_numbers = [x**2 for x in numbers]
print(squared_numbers)  # Output: [1, 4, 9, 16,
25]
```

Q.3 How do you filter elements in a list using list comprehension?

Ans: You can filter elements in a list using list comprehension by adding a conditional statement. For example, to filter even numbers:

```
numbers = [1, 2, 3, 4, 5]
even_numbers = [x for x in numbers if x % 2 == 0]
print(even_numbers)  # Output: [2, 4]
```

Q.4 Can you have multiple if conditions in list comprehension?

Ans: Yes, you can have multiple if conditions in list comprehension by chaining them using the and or or operators. For example:

```
numbers = [1, 2, 3, 4, 5]
filtered_numbers = [x for x in numbers if x % 2 == 0 and x > 2]
print(filtered_numbers)  # Output: [4]
```

Q.5 How do you create a dictionary using list comprehension in Python?

Ans: You can create a dictionary using list comprehension by specifying key-value pairs within curly braces `{}`. For example:

```
keys = ['a', 'b', 'c']
values = [1, 2, 3]
my_dict = {k: v for k, v in zip(keys, values)}
print(my_dict)  # Output: {'a': 1, 'b': 2, 'c': 3}
```

Topic 7

Libraries used for data science: Pandas, NumPy, Seaborn, Matplotlib

Q.1 What is a lambda function in Python?

Ans: A lambda function is an anonymous function defined using the **`lambda`** keyword. It is typically used for short, one-line functions. For example:

```
square = lambda x: x**2
print(square(3))  # Output: 9
```

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Ans: List comprehension is a concise way to create lists in Python based on existing lists or other iterables. It combines the creation of a new list with a loop and optional conditional statements. For example:

```python
numbers = [1, 2, 3, 4, 5]
squared_numbers = [x**2 for x in numbers]
print(squared_numbers)  # Output: [1, 4, 9, 16, 25]
```

Q.3 How do you filter elements in a list using list comprehension?

Ans: You can filter elements in a list using list comprehension by adding a conditional statement. For example, to filter even numbers:

```python
numbers = [1, 2, 3, 4, 5]
even_numbers = [x for x in numbers if x % 2 == 0]
print(even_numbers)  # Output: [2, 4]
```

Q.4 Can you have multiple if conditions in list comprehension?

Ans: Yes, you can have multiple if conditions in list comprehension by chaining them using the and or or operators. For example:

```python
numbers = [1, 2, 3, 4, 5]
filtered_numbers = [x for x in numbers if x % 2
== 0 and x > 2]
print(filtered_numbers)  # Output: [4]
```

Q.5 How do you create a scatter plot using Seaborn?

Ans: You can create a scatter plot using Seaborn's **`scatterplot()`** function, specifying the x and y variables from your dataset. For example:

```python
import seaborn as sns
import pandas as pd

df = pd.read_csv('data.csv')
sns.scatterplot(x='x_column', y='y_column',
data=df)
```

Topic 8

Types of plots in Seaborn and Matplotlib and their uses:

Q.1 What are some commonly used plots in Seaborn and Matplotlib?

Ans: Some commonly used plots in Seaborn and Matplotlib include:

- **Line plot:** shows the trend of a variable over time.
- **Scatter plot:** displays the relationship between two variables.
- **Bar plot:** compares categories or groups using rectangular bars.
- **Histogram:** visualizes the distribution of a continuous variable.
- **Box plot:** represents the distribution of a variable and displays outliers.
- **Heatmap:** shows the correlation or relationship between variables using colors.
- **Violin plot:** combines a box plot and a kernel density plot to represent the distribution of a variable.

Q.2 How do you create a box plot using Matplotlib?

Ans: You can create a box plot using Matplotlib's **`boxplot()`** function, providing the data and any additional parameters. For example:

```
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv('data.csv')
plt.boxplot(df['column'])
```

Q.3 How do you create a histogram using Seaborn?

Ans: You can create a histogram using Seaborn's **`distplot()`** function, specifying the variable and any additional parameters. For example:

```
import seaborn as sns
import pandas as pd

df = pd.read_csv('data.csv')
sns.distplot(df['column'])
```

Q.4 How do you create a bar plot using Matplotlib?

Ans: You can create a bar plot using Matplotlib's **`bar()`** or **`barh()`** functions, providing the data and any additional parameters. For example:

```
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv('data.csv')
plt.bar(df['x_column'], df['y_column'])
```

Q.5 How do you create a heatmap using Seaborn?

Ans: You can create a heatmap using Seaborn's **`heatmap()`** function, specifying the data, row and column variables, and any additional parameters. For example:

```
import seaborn as sns
import pandas as pd

df = pd.read_csv('data.csv')
sns.heatmap(data=df, x='x_column', y='y_column',
cmap='coolwarm')
```

Topic 9

Library for machine learning: Scikit-learn:

Q.1 What is Scikit-learn and how is it used in machine learning?

Ans: Scikit-learn is a popular machine learning library in Python that provides a wide range of algorithms and tools for various tasks such as classification, regression, clustering, dimensionality reduction, and model evaluation. It is widely used for building machine learning models and pipelines.

Q.2 How do you train a machine learning model using Scikit-learn?

Ans: To train a machine learning model using Scikit-learn, you typically follow these steps:

- Preprocess and prepare your data.
- Choose a suitable algorithm.
- Split your data into training and testing sets.
- Fit the model to the training data using the fit() method.

- Evaluate the model's performance using metrics and test data.

Q.3 How do you use cross-validation in Scikit-learn?

Ans: Scikit-learn provides the **`cross_val_score()`** function to perform cross-validation. You can specify the desired number of folds and the scoring metric to evaluate the model's performance. For example:

```
from sklearn.model_selection import
cross_val_score
from sklearn.linear_model import
LinearRegression

model = LinearRegression()
scores = cross_val_score(model, X, y, cv=5,
scoring='r2')
```

Q.4 How do you make predictions using a trained model in Scikit-learn?

Ans: Once you have trained a model in Scikit-learn, you can make predictions on new data using the **`predict()`** method. For example:

```
model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
```

Q.5 How do you save and load a trained model in Scikit-learn?

Ans: You can save a trained Scikit-learn model to disk using the **joblib** module's **dump()** function. To load a saved model, you can use the **load()** function. For example:

```
from sklearn.externals import joblib

# Save the model
joblib.dump(model, 'model.pkl')

# Load the model
loaded_model = joblib.load('model.pkl')
```

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